AXI_BUS
ADDR_LSB = 2
OPT_MEM_ADDR_BITS = 1
axi_awaddr[ADDR_LSB+OPT_MEM_ADDR_BITS:ADDR_LSB]

Custom_IP
ADDR_LSB = 2
WAWIDTH = 7
waddr_i_w = axi_awaddr[WAWIDTH+ADDR_LSB-1:ADDR_LSB];

Data	30-bit word addressable - alignment	Byte	
1f_a2_35_1c	xxxx_xxxx_xxxx_xxxx_xxxx_xxxx_xxxx_00_00	0x1c	axi_awaddr[3:2] = 2'h0
	xxxx_xxxx_xxxx_xxxx_xxxx_xxxx_xxxx_00_01	0x35	
	xxxx_xxxx_xxxx_xxxx_xxxx_xxxx_xxxx_00_10	0xa2	
	xxxx_xxxx_xxxx_xxxx_xxxx_xxxx_xxxx_00_11	0x1f	
00_01_7a_bf	xxxx_xxxx_xxxx_xxxx_xxxx_xxxx_xxxx_01_00	0xbf	axi_awaddr[3:2] = 2'h1
	xxxx_xxxx_xxxx_xxxx_xxxx_xxxx_xxxx_01_01	0x7a	
	xxxx_xxxx_xxxx_xxxx_xxxx_xxxx_xxxx_01_10	0x0a	
	xxxx_xxxx_xxxx_xxxx_xxxx_xxxx_xxxx_01_11	0x00	
23_cd_8a_3f	xxxx_xxxx_xxxx_xxxx_xxxx_xxxx_xxxx_10_00	0x23	axi_awaddr[3:2] = 2'h2
	xxxx_xxxx_xxxx_xxxx_xxxx_xxxx_xxxx_10_01	0xcd	
	xxxx_xxxx_xxxx_xxxx_xxxx_xxxx_xxxx_10_10	0x8a	
	xxxx_xxxx_xxxx_xxxx_xxxx_xxxx_xxxx_10_11	0x3f	
00_00_00_01	xxxx_xxxx_xxxx_xxxx_xxxx_xxxx_xxxx_11_00	0x01	axi_awaddr[3:2] = 2'h3
	xxxx_xxxx_xxxx_xxxx_xxxx_xxxx_xxxx_11_01	0x00	
	xxxx_xxxx_xxxx_xxxx_xxxx_xxxx_xxxx_11_10	0x00	
	xxxx_xxxx_xxxx_xxxx_xxxx_xxxx_xxxx_11_11	0x00	

raddr_i[2:0]
raddr_i[2] == 1
VO_MEM_BASE_ADDR

		AXI Mapping	
raddr_i[2:0] = 3'b100 = 4	axi_arddr[4:2] = 3'b{{1},{00}}	0x10	sudo devmem2 0xA0000010 w
raddr_i[2:0] = 3'b101 = 5	axi_arddr[4:2] = 3'b{{1},{01}}	0x14	sudo devmem2 0xA0000014 w
raddr_i[2:0] = 3'b110 = 6	axi_arddr[4:2] = 3'b{{1},{10}}	0x18	sudo devmem2 0xA0000018 w
raddr_i[2:0] = 3'b111 = 7	axi_arddr[4:2] = 3'b{{1},{11}}	0x1C	sudo devmem2 0xA000001C w

raddr_i[2] == 0 => {31'b{0}, complete_o}

AXI address

	Decimal		
waddr_i_w = axi_awaddr[8:2] = 7'b{{000},{0000}} LOAD_BASE_ADDR = 0	0	waddr_i_w = axi_awaddr[8:2] = 7'b{{000},{0000}}	sudo devmem2 0xa0000000 w 0x00000001
waddr_i_w = axi_awaddr[8:2] = 7'b{{001},{0000}} START_BASE_ADDR = 1	16	waddr_i_w = axi_awaddr[8:2] = 7'b{{001},{0000}}	sudo devmem2 0xa0000040 w 0x00000001
waddr_i_w = axi_awaddr[8:2] = 7'b{{010},{0000}} DONE_BASE_ADDR = 2	32	waddr_i_w = axi_awaddr[8:2] = 7'b{{010},{0000}}	sudo devmem2 0xA0000080 w 0x00000001
waddr_i_w = axi_awaddr[8:2] = 7'b{{011},{0000}} MATRIX_MEM_BASE_ADDR = 3	48	waddr_i_w = axi_awaddr[8:2] = 7'b{{011},{0000}}	sudo devmem2 0xA00000C0 w 0x00000221
		waddr_i_w = axi_awaddr[8:2] = 7'b{{011},{0001}}	sudo devmem2 0xA00000C4 w 0x00000376
		waddr_i_w = axi_awaddr[8:2] = 7'b{{011},{0010}}	sudo devmem2 0xA00000C8 w 0x0000017b
		waddr_i_w = axi_awaddr[8:2] = 7'b{{011},{0011}}	sudo devmem2 0xA00000CC w 0xffffe71
		waddr_i_w = axi_awaddr[8:2] = 7'b{{011},{0100}}	sudo devmem2 0xA00000D0 w 0x000001c4
		waddr_i_w = axi_awaddr[8:2] = 7'b{{011},{0101}}	sudo devmem2 0xA00000D4 w 0x00000317
		waddr_i_w = axi_awaddr[8:2] = 7'b{{011},{0110}}	sudo devmem2 0xA00000D8 w 0x00000124
		waddr_i_w = axi_awaddr[8:2] = 7'b{{011},{0111}}	sudo devmem2 0xA00000DC w 0xffffffea1
		waddr_i_w = axi_awaddr[8:2] = 7'b{{011},{1000}}	sudo devmem2 0xA00000E0 w 0xffffffffa
		waddr_i_w = axi_awaddr[8:2] = 7'b{{011},{1001}}	sudo devmem2 0xA00000E4 w 0xffffffdc5
		waddr_i_w = axi_awaddr[8:2] = 7'b{{011},{1010}}	sudo devmem2 0xA00000E8 w 0xffffffcc2
		waddr_i_w = axi_awaddr[8:2] = 7'b{{011},{1011}}	sudo devmem2 0xA00000EC w 0xffffffd14
		waddr_i_w = axi_awaddr[8:2] = 7'b{{011},{1100}}	sudo devmem2 0xA00000F0 w 0x000003d9
		waddr_i_w = axi_awaddr[8:2] = 7'b{{011},{1101}}	sudo devmem2 0xA00000F4 w 0x00000224
		waddr_i_w = axi_awaddr[8:2] = 7'b{{011},{1110}}	sudo devmem2 0xA00000F8 w 0xffffffde5
		waddr_i_w = axi_awaddr[8:2] = 7'b{{011},{1111}}	sudo devmem2 0xA00000FC w 0x000002a5
waddr_i_w = axi_awaddr[8:2] = 7'b{{100},{0000}} VI_MEM_BASE_ADDR = 4	64	waddr_i_w = axi_awaddr[8:2] = 7'b{{100},{0000}}	sudo devmem2 0xA0000100 w 0xffffff81
		waddr_i_w = axi_awaddr[8:2] = 7'b{{100},{0001}}	sudo devmem2 0xA0000104 w 0x00000133
		waddr_i_w = axi_awaddr[8:2] = 7'b{{100},{0010}}	sudo devmem2 0xA0000108 w 0x00000012
		waddr_i_w = axi_awaddr[8:2] = 7'b{{100},{0011}}	sudo devmem2 0xA000010C w 0x0000038d
			sudo devmem2 0xA0000040 w 0x00000001